Titration –I

(Link HCl)Vs (standard Na_2CO_3)

s.no	Volume of standard Na_2CO_3 (ml)	Burette readings		Concordant value (Volume of HCl) (ml)	Indicator
		Initial (ml)	Final (ml)		
1	20	0			Methyl orange
2	20	0			
3	20	0			

Calculation :

Volume of HCl (link) solution N_1 = ml

Normality HCl (link) solution N_1 = ? N

Volume of standard Na_2CO_3 solution N_2 = 20 ml

Normality of standard Na_2CO_3 solution N_2 = 0.0948 N

According to normality equation:

V_1

$$N_1 = \frac{V_2 \times N_2}{V_1}$$

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$

Normality HCl (link) solution _____ N

Titration–II

(Unknown NaOH) Vs (Link HCl)

Burette is washed with water, rinsed with HCl solution and filled with same HCl solution up to the zero mark. Exactly 20 ml of unknown NaOH solution is pipetted out into the clean, washed conical flask. To This solution 2 to 3 drops of phenolphthalein indicator is added and titrated against HCl link solution from the burette. HCl is added drop wise till the pink colour disappears completely. Burette reading is noted and the same procedure is repeated to get concordant values.

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